

Lab 7

ThreadHW.java

```
1- class FirstThread implements Runnable {
2-
3-     public void run() {
4-         try {
5-             for (int i = 0; i <= 9; i++) {
6-                 System.out.println("First Thread: " + i);
7-                 Thread.sleep(2000);
8-             }
9-         } catch (InterruptedException e) {
10-             System.out.println("First thread interrupted.");
11-         }
12-         System.out.println("First thread finished.");
13-     }
14- }
15-
16-
17- class SecondThread implements Runnable {
18-
19-     public void run() {
20-         try {
21-             for (int i = 0; i <= 9; i++) {
22-                 System.out.println("Second Thread: " + i);
23-                 Thread.sleep(4000);
24-             }
25-         } catch (InterruptedException e) {
26-             System.out.println("Second thread interrupted.");
27-         }
28-         System.out.println("Second thread finished.");
29-     }
30- }
31-
32-
33- public class ThreadHW {
34-
35-     public static void main(String[] args) {
36-
37-         Thread t1 = new Thread(new FirstThread(), "Thread-1");
38-         Thread t2 = new Thread(new SecondThread(), "Thread-2");
39-
40-         t1.setPriority(Thread.MAX_PRIORITY);
41-         t2.setPriority(Thread.MIN_PRIORITY);
42-
43-         t1.start();
44-         t2.start();
45-
46-         try {
47-             t1.join();
48-             t2.join();
49-         } catch (InterruptedException e) {
50-         }
51-
52-         System.out.println("Main thread finished.");
53-     }
54- }
55-
```

Lab 7

The OutPut:

```
Second Thread: 0
First Thread: 0
First Thread: 1
Second Thread: 1
First Thread: 2
First Thread: 3
Second Thread: 2
First Thread: 4
First Thread: 5
Second Thread: 3
First Thread: 6
First Thread: 7
Second Thread: 4
First Thread: 8
First Thread: 9
Second Thread: 5
First thread finished.
Second Thread: 6
Second Thread: 7
Second Thread: 8
Second Thread: 9
Second thread finished.
Main thread finished.
```

Lab 7

ThreadHW.java

```
ThreadHW.java :
1- class FirstThread implements Runnable {
2-
3-     public void run() {
4-         try {
5-             for (int i = 0; i <= 9; i++) {
6-                 System.out.println(Thread.currentThread().getName() + ": " + i);
7-                 Thread.sleep(2000);
8-             }
9-         } catch (InterruptedException e) {
10-             System.out.println("First thread interrupted.");
11-         }
12-         System.out.println(Thread.currentThread().getName() + " finished.");
13-     }
14- }
15-
16-
17- class SecondThread implements Runnable {
18-
19-     public void run() {
20-         try {
21-             for (int i = 0; i <= 9; i++) {
22-                 System.out.println(Thread.currentThread().getName() + ": " + i);
23-                 Thread.sleep(4000);
24-             }
25-         } catch (InterruptedException e) {
26-             System.out.println("Second thread interrupted.");
27-         }
28-         System.out.println(Thread.currentThread().getName() + " finished.");
29-     }
30- }
31-
32-
33- public class ThreadHW {
34-
35-     public static void main(String[] args) {
36-
37-         Thread t1 = new Thread(new FirstThread(), "First Thread");
38-         Thread t2 = new Thread(new SecondThread(), "Second Thread");
39-
40-         t1.setPriority(Thread.MAX_PRIORITY);
41-         t2.setPriority(Thread.MIN_PRIORITY);
42-
43-         t1.start();
44-         t2.start();
45-
46-         try {
47-             t1.join();
48-             t2.join();
49-         } catch (InterruptedException e) {
50-             System.out.println("Main thread interrupted.");
51-         }
52-
53-         System.out.println("Main thread finished after both threads end.");
54-     }
55- }
```

Lab 7

The OutPut:

```
Second Thread: 0
First Thread: 0
First Thread: 1
Second Thread: 1
First Thread: 2
First Thread: 3
Second Thread: 2
First Thread: 4
First Thread: 5
Second Thread: 3
First Thread: 6
First Thread: 7
Second Thread: 4
First Thread: 8
First Thread: 9
Second Thread: 5
First Thread finished.
Second Thread: 6
Second Thread: 7
Second Thread: 8
Second Thread: 9
Second Thread finished.
Main thread finished after both threads end.

...Program finished with exit code 0
Press ENTER to exit console.□
```